Unit 9 - Computer Networks
Assignment 2 - Design a LAN Tasks 3&4

Introduction

Scenario / Vocational Context - *You are a junior consultant for T5 Alliance. A new client, Competition Bikes Ltd, are a new start-up company that manufacture a range of carbon fibre race and hybrid bikes from their small factory and sell them online. They have commissioned you to design and implement a new local area network with the following requirements:*

The network must support at least 100 client computers with room for 20% growth.
There are five departments: Management (8 staff), Accounting (4 staff), Engineering (70 staff), Sales & Marketing (10 staff) and IT (8).

The network topology should be designed to be fault tolerant and adhere to the "Core, Distribution, Access" model of network design.

5 LAN servers will connect to the master switch: 2 Domain Controllers, 1 File & Print Server, 1 Email server and 1 SQL Database Server

Network media should be provided as follows: Cat 6 UTP cable should connect the servers and PCs to the department switches. 1000BASE-SX fibre optic cables should connect the switches together.

There should be a DMZ that contains a further 3 servers: Web Server, FTP Server and a Remote Access Server.

The DMZ should provide the maximum security possible. Support for remote access must be included in your design.

Task 3

(U5P1) - Describe network technologies / (U9P3) - Explain the key components required for client workstations to connect to a network and access network resources

Task description - *All items that appear in your topology diagram MUST appear in the inventory with a description and a cost per unit. Total the cost of each component for an overall total price for the new network.*

Introduction

With the creation of the network topology diagram, the budget for the said diagram would need to fit with the hardware requirements to construct the network for office use. With a budget of £75,000, the hardware needed for the network would have to be listed with a description for its purpose, use and a cost for each of the units of hardware.

Inventory table

ITEM	DESCRIPTION/PURPOSE	UNIT COST	COST OF THE UNITS
Switch - *Cisco SG220-26P 26-Port Gigabit PoE Smart Plus Switch*	These switches would be used as client switches. The purpose for these switches is to connect the PCs, used by the staff, to the network itself. There will be 7 of these devices installed within the network.	7x £350	£2,450
Master Switch - *Cisco 300 Series Switch SG300-10*	These switches would be used as the master switch within the network. Basically, as the distribution layer of the network, so the routing and filtering is managed by these switches. There will be two of these devices connected to the network.	2x £136	£272
Firewall 1 - *SONICWALL TZ300 APPLIANCE*	This firewall switch will be used as the main firewall to the entire network. Essentially, this firewall switch will be used to control the connections to the network, including outgoing and incoming connections. This is so there is no unauthorised interception into the network. There will only be one of this device connected to the network.	1x £395	£395
Firewall 2 - *WatchGuard XTM 330*	This firewall switch will be used as the secondary firewall to the entire network. Basically, this firewall switch also will be used to control the connections to the network,	1x £684	£684

	but only when the first firewall has been breached of ceased to work. This is so there is more security to prevent unauthorised interception into the network. There will only be one of this device connected to the network.		
DMZ switch - *Cisco 300 Series Switch SG300-10*	This switch would be used as the DMZ of the network. Basically, this switch will be used to create more security to the network, by making a DMZ for extra security from incoming and outgoing connections. There will only be one of this device connected to the network.	1x £136	£136
Server - Dell PowerEdge FC630 *Rack Server*	These servers would be used for multiple purposes, which depends on what type of software is installed and where it is connected. The servers can be used as Domain Controllers, FTP, File and Print, Email, RAS, Web and as an SQL Database server. The reason that so much has been invested on these servers, is due to the multipurpose capability of these servers. Basically, they have been designed to have versatile installation both hardware and software wise, meaning that they can be installed around the	8x £7,505	£60,040

	network with a lot of different types of software.		
Domain Controllers	Domain controllers are usually built into Windows Server Operating Systems, so no extra installation is needed. The purpose for the domain controllers is to respond to security authentication requests. Basically, it acts as a guard to the server, letting in only requests which have been confirmed by the domain controller itself.		
FTP Server	The File Transfer Protocol is a software which is usually installed within a Server operating system. This type of software is used for transferring files around the network, usually directly from the server itself. Basically, this software moves files from and to the server.		
File & Print Server	A file and print server is typically used to connect printers within the network, giving the ability to manage the jobs of a printer. This type of software seems simple enough, due to the fact that it stores files within the server for the entire network, so they can be accessed by anyone on the network either digitally or physically.		

Email Server	Usually, a server operating system already has an email application installed within it. The purpose for this software is that it controls the emails that is sent and received on the network, also giving the ability to store the emails so that they can be read at any time.		
Remote Access Server	A remote access server's purpose is to provide routing and other forms of remote access to the network itself. The purpose of this is to make sure that the incoming and outgoing traffic is directed properly. Usually, it is used to allow the staff to connect to the entire network with their own devices.		
Web Server	A web server would need to have some sort of software installed, such as Apache HTTP or Nginx, which would give the ability to host websites within the storage of the server. The purpose of this server is to give the users within the network the accessibility of the world wide web, which can prove to be very advantageous.		
SQL Database Server	This server has software which gives the server the ability to store and manage information within the		

	server's storage. Basically, this type of server is used to take care of the data that is important to the clients within the network.		
Cat6 UTP LSOH Solid Core Cable 100mt	This cable is a Cat6 twisted pair cable which measures 100 meters in size. This cable is extremely common and has a good traffic speed, making it a good choice in connecting a network physically. The purpose of this cable is going to be to connect most of the network together, apart from the places taken by the fibre optic, so the network can carry out its intended purpose.	111x £48	£5328
Broadcom Black, 100m, Single Mode Fibre Optic Cable 1 Core	This cable is a fibre optic cable which measures 100 meters in size. This cable is less common than the twisted pair in comparison, but it is still has common use in network. This is due to the fact that fibre optic is significantly faster than twister pair, but it is more fragile and expensive to produce and purchase. Which is why I only used it to connect the switches, which the clients are connected to, to the distribution layers.	12x £57	£684
Cisco GLC-SX-MM	This transceiver is used for the fibre optic cables to	24x £4	£96

Compatible 1000BASE-SX SFP 850nm 550m Transceiver	connect to the devices themselves. The reason I chose a 1000BASE-SX transceiver is that the speed is very high but the length is limited to 100 meters, but distance isn't a problem since the office isn't extremely large.		
Operating System - Microsoft Window 10 (64Bit)	This piece of software is an OS made by Microsoft. This operating system is the latest made by this company and it is also the most popular, giving it massive versatility with hardware and software. It also has very good user friendliness, meaning that amateurs could use it without significant issues.	*Price of software included with the user's computers.*	*Price of software included with the user's computers*
Server Operating System - Microsoft Windows Server 2016	This piece of software is a server OS also made by the company Microsoft. This operating system contains a wide amount of applications and other forms of software features which allows a server to be setup to most the common jobs for a server. Basically, this software allows servers to carry out their intended purpose through this versatile piece of software.	*Price of software included with the servers.*	*Price of software included with the servers.*
Budget - £75,000 Total cost - £70,085 including VAT			

Task 4

(U9P4) - Explain the function of interconnection devices / (U10P1) - Explain how networks communicate

Task description - *Explain network cards, hubs, layer 2 switches and routers. Each device must be described in terms of its function, roughly how it works and how "intelligent" it is (including for example, how it makes decisions about where to send frames/packets) and the layer of the OSI model is exists at.*

Introduction

Obviously, network devices play a significant role when it comes to network management, due to the fact that the network itself would not be of much use without network devices to run from. Having a lack of these devices would mean that the network would be impossible to manage, making the traffic an uncontrollable mess.

These devices are measured using the OSI model, which determines how intelligent the device is and how it works within the network. The OSI model consists of: Physical (First layer), Data Link (Second layer), Network (Third layer), Transport (Fourth layer), Session (Fifth layer), Presentation (Sixth layer) and Application (Seventh layer).

In very simple terms, data cannot reach the other side without something to take it there, which is where network devices come in. However, not all network devices are the same. For instance, there are many forms of network devices, each made for a specific but useful purpose. These network devices consist of: network interface cards, hubs, routers and switches.

Network interface cards

Network interface controllers are also known as LAN Adaptors or Network Interface Cards, and are small pieces of hardware that fits inside of the computer or laptop itself.

On the OSI model, a network card operates at the layer 1 section, making this device a very unintelligent piece of hardware. The reason it is a layer 1 is due to the fact that network cards do not make decisions based on the MAC address, instead just sending strings of binary data down the cables.

To specify, network cards implement a communication between the network and the computer in order for it to connect to the internet. Network cards can come with either a wired Ethernet port or can connected to wifi through a wireless network card. An example of this would be college computers which are connected to the LAN through wired network cards.

Overall, network cards are simple and unintelligent pieces of hardware which are situated within the first layer of the OSI model. The reason network cards are so simple in design is due to their simple design purpose, which is to send binary data from one end to another.

Hubs

A hub, commonly known as a connection point for a computer network, is a simple network device which operates at the Layer 1 of the OSI model, due to the fact that when one device sends out a message to another device, the hub does not decide where the message goes.

Since hubs work at the first or Physical layer of the OSI model, it links all the devices connected to it and forms a single network, sometimes it even transmits data to each port connected to the network.

However, there are three kinds of Hubs which are: A passive hub, which enables the packets to go from one port to the other, an “intelligent” hub, which allows admins to monitor traffic and configure each port and a switching hub, which can read the destination of each packets and send it to the right port.

When in use, it repeats the message to all the ports, meaning that each device that directly connects to the hub uses a port on the hub. However, this can cause some disadvantages to using a hub. For example, one disadvantages is that the all traffic on one port is copied to the corresponding ports, due to the fact that this device is only a layer 1 device.

This causes a problem with hubs, which is that they share bandwidth among everyone. Metaphorically, a telephone party line is like a hub. Each person decides by the ring if it is their phone call or someone else’s call, BUT, everyone can listen in on the call.

Another limitation to hubs is that the data, which is sent to the network, would have a slow traffic speed, which isn’t a disadvantage in itself, but also when the data can’t be seen unless the previous data traffic is sent means that you can only see the data when it arrives. However, the LAN can see the packet because it allows the packets of data to be transferred from one port to another.

Overall, a hub is also a very simple network device, due to it being a layer 1 device. However, it is for the same reason as the previous layer 1 device, it's due to the simple job the network device has. Which is to send messages to all of the ports that it is connected to within the network.

Layer 2 switches

Switch is a device that makes connection between network devices by creating network segments, they are essentially a device which links together other network devices. Metaphorically, a private phone call is like a switch. The phone number that is entered is looked up in the table and the correct telephone rings at the other end.

Basically, a switch could be seen as a less “intelligent” router. However, this doesn't mean that there aren’t different types of switches used for different purpose and causes.

Actually, they are very versatile that, in the OSI model, switches don’t only operate at layer 2. Meaning that it is possible to obtain a more intelligent layer 3 switch, which can carry out the same purposes, but to a much more sophisticated level.

When being used, switches increase network bandwidth, reduces work on single computer and increases network performance as the switch will switch between computer so that all the computers connected to that network will get fair amount of work to do and handle with.

Usually, Switches link together the different devices usually by an Ethernet port, helping to send packets between the devices whilst helping the, to communicate with one another. Switches operate at the Data Link or second layer. Making, switches “intelligent” hubs but less “intelligent” routers.

One advantage that switches have, is that switches can remember which ports are connected to which devices. Meaning, when a switch receives a packet (data), it resends that packet directly to the correct port.

For example, if host A sends out a message through port A. The switch records into its switch table that host A is on port A. For example, when host B decides to send a packet to host A, the switch first checks its switch table. If port A is registered in the switch table, it will resend the packet directly to port A instead of sending it to all the ports, which also means that switching gives dedicated bandwidth.

Overall, a switch is more intelligent than any of the layer 2 devices, such as a hub and a network card, due to its intelligence when handling and sending data to specific hosts and ports. Sometimes, a switch could be more intelligent, which would be a layer 3 switch, which carries out more sophisticated data handling when compared to a layer 2 switch.

Routers

A router is a small but complex physical device which has the job of connecting multiple networks together into the same place, meaning that it is a networking device that forwards data packets between connected networks.

Basically, it is made specifically for receiving, analysing and sending packets to another network interface, which creates an inter-network for the computers to connect. Meaning that routers, in a sense, are more “intelligent”, as they are capable of analysing the data being sent over a network. Which is why routers are found on the third layer of the OSI model, which is the Network layer.

Routers are commonly used in a home or office so share a single internet connection between multiple computers and devices, usually connected to two or more data lines from different networks. In the home or office, the user can connect to the router via cable or they can simply connect to it via wireless, or Wi-Fi.

Using a network protocol, like TCP/IP, a router can intelligently move data from one network to another. Which is why, due to the intelligence of the router, that the entire Internet consists of many interconnected routers.

When being used, routers passes data such as email, video, pictures, etc between the computers. Hence why routers are called routers, which is due to the fact it routes data from and to where it needs to go. Essentially, when data comes in from one of the line the router reads the address to finalise the data's location.

For example, when a user sends a request for a server, the router in the local network will check its routing table and decide where to resend the data. So, a router needs to have a better understanding of the whole network structure than does a layer 2 switch.

Metaphorically, a long distance phone call uses a router like device. The initial one plus area code tells the system that the call is not local but needs to be routed to a distant phone network. The "phone router" then connects the call to the correct distant phone network.

Overall, a router is a very complex network device when compared to other network devices such as switches, hubs and network cards. This is due to the most more heavy duty and complexity that routers are intended to carry out when compared to these devices, which is why such large networks such as the World Wide Web and other large forms of networks contains routers almost all the time.

Conclusion

In conclusion, Layer 1 devices, such as hubs and network cards, aren't very "intelligent" when compared to the devices ranked in the 2 layers above them. Due to the fact they are made to carry out simple jobs, hence why most layer 1 devices are very simple in design and operation.

However, this does not render the layer 1 devices less useful than the layer 2 or layer 3 devices. The reason for this is because some network could just be too small to use layer 2 or layer 3 devices, which would be overkill to use on a small network. Also, if a network did not consist of layer 1 devices, it would most likely not work very well, but larger networks can be an exception.

Layer 2 devices, such as common switches, are more "intelligent" when compared to layer 1 devices, but not very "intelligent" when compared to layer 3 devices. This is due to the fact that they are made to carry out jobs which are more complex than a layer 1 operation, but a less complex job than a layer 3 device would operate.

Again, this does not mean that it is useless, in fact it is debatably more useful than a layer 1 device such as a hub. Due to the fact that a switch is commonly regarded as an “intelligent” hub. Which is why, on larger networks, it is more common to find a switch instead of a hub. Which means that a layer 2 device is more important than a layer 1 device, but it still doesn’t render layer 1 completely useless.

Finally, layer 3 devices, such as routers and layer 3 switches, are even more “intelligent” when compared to layer 2 and a lot more “intelligent” when compared to layer 1. Meaning that layer 3 devices were designed to be complex devices to carry out complex operations within the network, to improve the traffic on the entire network or even traffic to external networks.

However, this again does not render the less “intelligent” devices useless because they carry out specific jobs, while more “intelligent” can carry out more jobs but not those specific jobs that layer 2 and layer 1 devices carry out. Basically, it is important to have a variety of devices carrying out different jobs, instead of having very few devices carrying out many jobs. Which can cause performance and management problems.

Overall, it does not matter how “intelligent” or what layer on the OSI model the devices are in, they all are included within the network for a specific reason. However, there are some devices which can be replaced, but this does not mean that all devices can be replaced.

This is due to how complex the network is and how it communicates with itself and with other external networks, such as connecting a LAN to a WAN using a router or a client connecting to a switch through a network card and cable. So, both simple and complex jobs are needed to make a network carry out traffic in an efficient manner, instead of making an intelligent device carry out all the operations by itself.